

Artur Goncharov

Backend / Full-stack engineer · HVAC engineer

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Flagship: [wentmarket.ru](#) · Side: [github.com/goncharovart/fan-selector-api](#)

SUMMARY

Backend-leaning full-stack engineer with **6 years in HVAC engineering software** and ~2 years shipping production web stacks. I work spec-first with AI coding agents (Claude Code, Cursor) end-to-end — spec → architecture → implementation → deploy. My niche is domains where the algorithm matters as much as the plumbing: fan-curve matching, duct loss, life-cycle cost, acoustic selection.

Comfortable in TypeScript/Node primary stack; currently ramping on Go via a real side project to broaden the backend toolbelt.

SELECTED EXPERIENCE

Wentmarket — Founder / Lead engineer · 2024–present

B2B platform for HVAC engineers, in production at [wentmarket.ru](#). Built solo with AI-assisted workflow (Claude Code primary). Real users, real orders, real payments.

- **Fan-selection engine.** Indexed 17 000+ polynomial performance curves in PostgreSQL with GiST range filtering; duty-point matching solves $P_{fan}(Q) = P_{target}$ per request and returns ranked results in <100 ms p95.
- **Order pipeline.** Server actions + BullMQ workers + Redis L2 cache. Yookassa payment integration with deterministic SHA-256 idempotency keys and 54-FZ fiscal receipts. Webhook validation via API call rather than HMAC to defeat replay.
- **CDEK delivery integration.** 871-line service layer: OAuth token refresh on 401, exponential backoff (3 retries, max 8 s), 30 s per-request timeout, full order/intake/tracking/waybill flow.
- **Soft-delete extension** via Prisma `$extends` across 8 models with consistent query filtering.
- **Headless Bitrix24 loader** — loads the vendor widget, hides its UI via injected CSS + MutationObserver, exposes `window.openB24Chat()` for our own triggers. Avoids two competing chat buttons.
- **1C ERP sync** through CommerceML XML; products/prices/stock pipeline.
- **Meilisearch** with graceful fallback to Prisma `LIKE` when the indexer is unreachable — single observable degradation rather than a crash.
- **Stack:** Next.js 16, React 19, TypeScript, Prisma 7, PostgreSQL, Redis (BullMQ + cache), Meilisearch, Sentry, NextAuth v5.
- **Infra:** self-hosted VPS, systemd + nginx, custom deploy pipeline (`tar` | `ssh` | `build-on-server`).

Goal-Energo — HVAC project engineer · 2018–2024

Six years sizing ventilation, smoke-extract, and AHU systems for industrial and commercial buildings. Built internal sizing spreadsheets/scripts that later became the seed for Wentmarket's engineering engines.

- Polynomial curve fitting from measured fan data → coefficient catalog used today in Wentmarket.
- Authored project sizing methodology used across the team.
- Customer-facing: gathered duty-point requirements, produced specifications and bid packages.

SIDE PROJECT — FAN-SELECTOR-API

github.com/goncharovart/fan-selector-api

A standalone Go microservice that extracts Wentmarket's fan-matching engine into a clean, testable service. Built spec-first (BMad-style). **Purpose:** demonstrate Go + cloud-deployable architecture + SDD workflow on a domain I know cold.

- **Go 1.23**, chi router, pgx/v5, go-redis/v9, OpenTelemetry, slog.
- **Postgres 16** with GiST range index on the fan duty-point envelope; prefilter narrows from full catalog to candidates in a single indexed scan.
- **Bisection root finder** for $P_{\text{fan}}(Q) = P_{\text{target}}$ with input defenses (NaN/Inf, sign-change check, boundary hits, monotone-decreasing curve assumption documented). Benchmarked: 0.65 ns Horner eval, 53 ns bisection, 3.7 μ s for a 50-candidate Evaluate pass — 99 ms of headroom inside the 100 ms p95 target.
- **Redis cache** with deterministic SHA-256 keys, TTL jitter ($\pm 10\%$) to prevent stampedes, graceful degradation to a NopCache when Redis is down.
- **Distroless multi-stage Docker image**, non-root user.
- **Two deployment paths in repo** — `cloudbuild.yaml` for GCP Cloud Run (Artifact Registry + Cloud SQL + Secret Manager) and `fly.toml` for Fly.io; both fully written, ready to `apply` once a billing account is attached.
- **Embedded SQL migrations** — `cmd/migrate` ships in the same distroless image as the server, applies migrations on every release via Fly's `release_command` or Cloud Run's pre-start hook.
- **OpenTelemetry tracing** with manual spans across `cache.get / cache.set`, `storage.candidates`, `match.evaluate`; per-span attributes for cache hit, candidate counts, value bytes, TTL. Exporter ships to any OTLP collector.
- **Integration tests** via testcontainers-go: every CI run boots a real Postgres 16 container, applies embedded migrations, asserts the GiST range prefilter narrows the catalog correctly. Unit + integration + benchmark jobs run in parallel.
- **GitHub Actions CI:** `go vet`, `go test -race`, golangci-lint, integration suite, benchmark report, image build.
- **SDD artifacts:** PRD, architecture, per-story acceptance criteria ([specs/](#)) written before implementation. Each story closes with a self-contained PR.

Runs locally via `docker compose up -d && go run ./cmd/server`.

SKILLS

Backend. TypeScript/Node, Go (ramping), PostgreSQL, Redis, BullMQ, REST, OpenAPI, OpenTelemetry, structured logging, idempotency, observability.

Cloud / DevOps. Docker (multi-stage, distroless); GitHub Actions CI/CD; Linux/systemd; nginx; custom Bash deploy pipelines (`tar` | `ssh` | `build-on-server`). Deployment configs written for GCP Cloud Run (Cloud Build / Cloud SQL / Secret Manager) and Fly.io; familiarity at the config-and-docs level, not yet hands-on production.

AI-assisted workflow. Spec-driven development (BMad-Method, custom spec templates), Claude Code as primary IDE companion, subagent parallelization for independent tasks, automated test loops, prompt-based hooks for guardrails.

Frontend. Next.js, React, TypeScript, Tailwind, Framer Motion, Prisma.

Domain. HVAC system sizing, fan / duct / acoustic / life-cycle-cost calculations, project specification, 1C ERP integration, CDEK logistics, Yookassa / 54-FZ fiscal compliance.

EDUCATION

Don State Technical University — engineering degree, Heating, Ventilation and Air-Conditioning. 2013–2018.

LANGUAGES

Russian — native · English — B2 (technical reading/writing fluent; speaking improves under load).

NOTES FOR THE RECRUITER

- **Go experience is recent.** Strong in TypeScript backend, currently building a real Go service to close the gap (see `fan-selector-api`). Honest about the ramp; not honest about pretending it's not a ramp.
- **AI-assisted shipping is my baseline, not an experiment.** Wentmarket shipped to production with Claude Code at every step — PRDs, architecture, per-feature stories, code, tests, deploy. I can talk through any commit.
- **Open to relocation** to Poland / Georgia / Belarus / other.
- Available immediately.